

A PROJECT REPORT ON
FINANCIAL RISK MODELLING

Codtech IT Solutions Internship

Task 1: Financial Risk Modelling

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Domain: Data Analytics

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1. Introduction

Financial Risk Modelling is the process of using statistical techniques, data analysis, and machine learning algorithms to identify, assess, and predict potential financial risks faced by individuals, organizations, or financial institutions. These risks may include loan defaults, credit failures, market fluctuations, investment losses, and other uncertainties that can affect financial stability.

In the banking and financial sector, risk assessment plays a crucial role in decision-making. Financial institutions must evaluate the creditworthiness of customers before approving loans or credit facilities. Traditional risk assessment methods often rely on manual analysis and predefined rules, which may be time-consuming and prone to errors. With the growth of data science and artificial intelligence, machine learning techniques have become powerful tools for predicting financial risk more accurately and efficiently.

This project focuses on developing a Financial Risk Modelling system using machine learning. The system analyzes customer financial information such as income, employment length, loan amount, interest rate, loan-to-income ratio, and credit history to determine whether an applicant is likely to be a low-risk or high-risk borrower. Various data preprocessing techniques, exploratory data analysis, visualization methods, and machine learning algorithms are applied to extract meaningful insights and improve prediction accuracy.

The project also includes the development of an interactive dashboard that enables users to input customer details and obtain real-time risk predictions. Such systems help financial institutions make informed decisions, reduce potential losses, improve operational efficiency, and support responsible lending practices.

Overall, Financial Risk Modelling demonstrates how data-driven approaches and machine learning can enhance risk management processes and contribute to more effective financial decision-making.

2. Importance of Financial Risk Modelling

Financial Risk Modelling is an essential tool used by banks, financial institutions, insurance companies, and investment firms to identify, measure, and manage potential financial risks. It helps organizations make informed decisions by analyzing historical and current financial data to predict future outcomes.

1. Reduces Financial Losses

Financial risk models help identify high-risk customers and transactions before losses occur. This enables institutions to take preventive actions and minimize financial damage.

2. Improves Loan Approval Decisions

Banks and lending institutions use risk modelling to evaluate the creditworthiness of loan applicants. This helps ensure that loans are granted to customers who are capable of repaying them.

3. Enhances Decision-Making

By providing accurate risk assessments, financial risk modelling supports better strategic and operational decisions related to lending, investments, and resource allocation.

4. Helps in Fraud Detection

Risk models can identify unusual patterns and suspicious activities that may indicate fraud, helping organizations protect their assets and customers.

5. Improves Customer Segmentation

Financial institutions can classify customers into different risk categories, enabling them to offer personalized financial products and services.

6. Assists in Investment Planning

Investors and financial analysts use risk modelling to evaluate potential investment risks and optimize portfolio performance.

7. Supports Regulatory Compliance

Financial institutions must comply with various government and regulatory guidelines. Risk modelling helps organizations meet these requirements by maintaining proper risk management practices.

8. Promotes Financial Stability

Effective risk management contributes to the stability and sustainability of financial institutions, reducing the likelihood of defaults and financial crises.

3. Dataset Description

The dataset used in this project is a Credit Risk Dataset obtained from a publicly available source. The dataset contains information about loan applicants and their financial profiles, which is used to predict whether a customer is likely to be a low-risk or high-risk borrower.

The primary objective of the dataset is to assist financial institutions in evaluating loan applications and identifying potential loan defaults. By analyzing customer demographic information, employment details, loan characteristics, and credit history, machine learning models can estimate the level of financial risk associated with each applicant.

3.1 Dataset Features

The dataset contains the following important attributes:

Attribute	Description
person_age	Age of the loan applicant
person_income	Annual income of the applicant
person_home_ownership	Home ownership status of the applicant
person_emp_length	Number of years the applicant has been employed
loan_intent	Purpose of the loan (education, medical, personal, etc.)
loan_grade	Credit grade assigned to the loan
loan_amnt	Amount of loan requested
loan_int_rate	Interest rate applied to the loan

loan_percent_income	Ratio of loan amount to annual income
cb_person_default_on_file	Previous loan default history
cb_person_cred_hist_length	Length of the applicant's credit history
loan_status	Target variable indicating loan default risk

3.2 Target Variable

The target variable used for prediction is:

loan_status

- 0 = Low Risk / Loan Approved
- 1 = High Risk / Loan Default Risk

The machine learning models are trained to classify applicants into these categories based on their financial and credit-related information.

3.3 Dataset Characteristics

- Contains customer financial and demographic information.
- Includes both numerical and categorical features.
- Suitable for classification problems.
- Used for credit risk assessment and loan default prediction.
- Requires data preprocessing, encoding of categorical variables, and feature scaling before model training.

3.4 Purpose of the Dataset

The dataset helps in understanding the factors that influence loan repayment behavior and enables the development of predictive models for financial risk assessment. Such models assist banks and financial institutions in reducing loan defaults, improving credit decisions, and enhancing overall risk management processes.

4. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is an important step in the machine learning process. It involves examining, summarizing, and visualizing the dataset to understand its structure, identify patterns, detect anomalies, and discover relationships among variables. In this project, EDA was performed to gain insights into the factors influencing loan default risk.

4.1 Data Overview

The dataset was loaded and inspected to understand the number of records, features, data types, and target variable distribution. Statistical summaries such as mean, median, minimum, maximum, and standard deviation were generated for numerical attributes.

4.2 Distribution of Loan Status

A count plot was used to visualize the distribution of loan status categories. The analysis showed the proportion of low-risk and high-risk borrowers in the dataset. This helped determine whether the dataset was balanced or imbalanced.

4.3 Age Distribution Analysis

A histogram was used to analyze the age distribution of loan applicants. The majority of applicants were concentrated in the young and middle-age groups, indicating that most borrowers were economically active individuals.

4.4 Income Distribution Analysis

Income distribution was visualized using histograms and density plots. Applicant incomes varied significantly, with a majority of customers belonging to the middle-income group. A small number of applicants had exceptionally high incomes.

4.5 Loan Amount Analysis

The loan amount distribution was examined to understand borrowing patterns. Most applicants requested moderate loan amounts, while fewer customers applied for large loans.

Fig 4.1. Heatmap

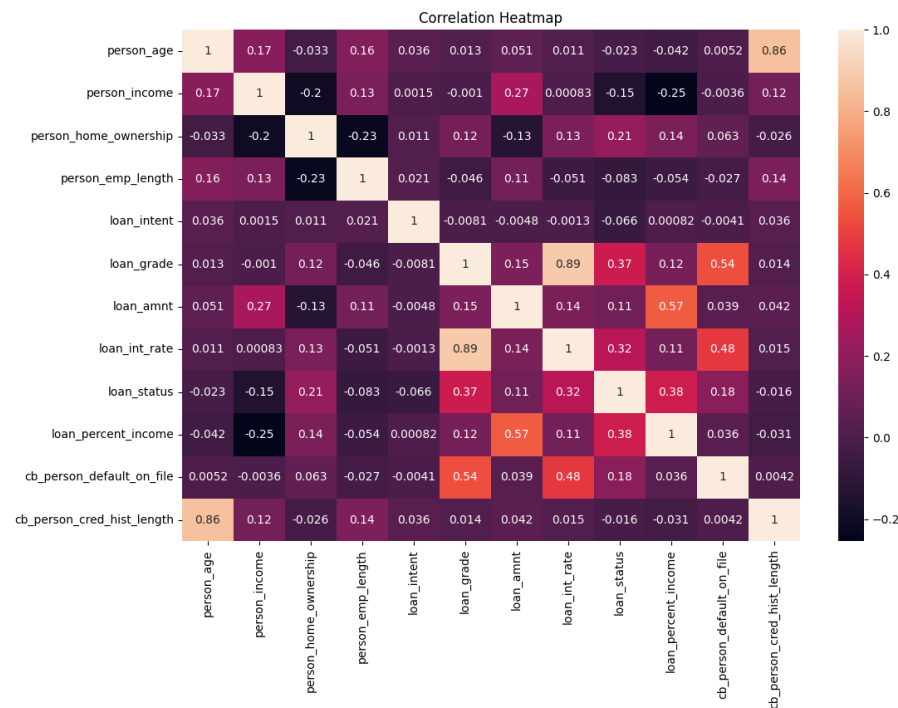
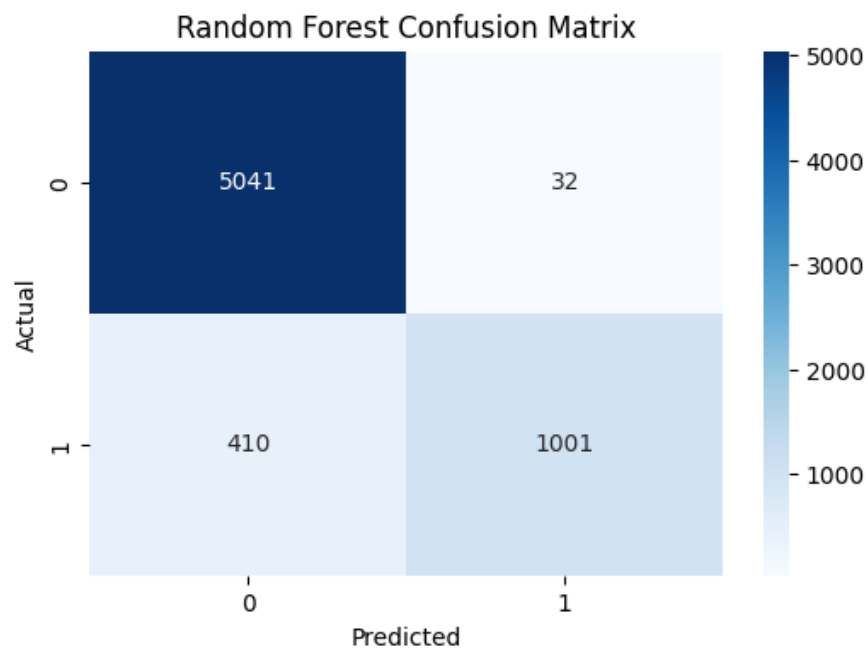


Fig 4 2. Confusion Matrix



5. Results and Findings

After performing data preprocessing, exploratory data analysis, feature engineering, and model training, multiple machine learning algorithms were evaluated to determine the most effective model for financial risk prediction.

5.1 Model Performance Comparison

The performance of different classification models was evaluated using accuracy as the primary metric.

Model	Accuracy
Logistic Regression	84.19%
Decision Tree	88.46%

Among the evaluated models, the Decision Tree Classifier achieved the highest accuracy of 88.46%, outperforming Logistic Regression. Therefore, the Decision Tree model was selected as the final model for financial risk prediction.

5.2 Confusion Matrix Analysis

A confusion matrix was generated to evaluate the classification performance of the selected model. The matrix provided information about correctly classified and misclassified loan applications. The results showed that the model was able to correctly identify the majority of low-risk and high-risk applicants, demonstrating strong predictive capability.

5.3 Dashboard Results

The dashboard successfully integrated the trained machine learning model and preprocessing components, ensuring accurate and consistent predictions. The prediction process is completed within seconds, making the system efficient and suitable for practical applications.

The Streamlit dashboard transformed the machine learning model into a usable application, allowing users to interact with the financial risk prediction system in a simple and intuitive manner. The dashboard serves as a proof-of-concept for how artificial intelligence and data analytics can be applied in modern financial institutions to support credit risk management and lending decisions.

6. Dashboard Development

To make the financial risk prediction system interactive and user-friendly, a web-based dashboard was developed using the Streamlit framework. The dashboard serves as an interface between the user and the machine learning model, allowing real-time financial risk assessment without requiring technical knowledge of machine learning.

6.1 Purpose of the Dashboard

The primary purpose of the dashboard is to provide an easy-to-use platform where users can enter customer financial information and instantly receive a risk prediction. The dashboard converts the trained machine learning model into a practical application that can be used by financial institutions for decision-making.

6.2 Dashboard Architecture

The dashboard consists of three main components:

1. User Input Module
2. Machine Learning Prediction Module
3. Visualization and Results Module

The user enters financial information through input fields. The entered data is processed using the same preprocessing techniques applied during model training. The trained machine learning model then generates a prediction, and the result is displayed on the dashboard.

6.3 Dashboard Features

The developed dashboard includes the following features:

Users can enter important financial details such as:

- Age
- Annual Income
- Employment Length
- Loan Amount
- Interest Rate
- Loan Percentage Income
- Credit History Length

Risk Prediction

The dashboard uses the trained machine learning model to predict whether the applicant belongs to:

- Low Risk Category
- High Risk Category

Data Visualization

The dashboard displays various charts generated during Exploratory Data Analysis, including:

- Age Distribution
- Income Distribution
- Loan Distribution
- Correlation Heatmap
- Confusion Matrix

These visualizations help users understand the dataset and the factors influencing financial risk.

Interactive Interface

The Streamlit framework provides an interactive and responsive user interface, enabling users to perform risk assessment efficiently.

6.4 Dashboard Workflow

The working process of the dashboard is as follows:

1. User enters financial details.
2. Input data is collected by the dashboard.
3. Data is preprocessed using the saved scaler.
4. The trained machine learning model receives the processed data.
5. The model predicts the applicant's risk category.
6. The prediction result is displayed instantly.
7. Users can view graphs and charts for additional insights.

6.5 Dashboard Testing

The dashboard was tested using multiple sample records from the dataset. Different combinations of income, loan amount, employment length, and credit history were entered to verify the prediction accuracy and functionality of the system.

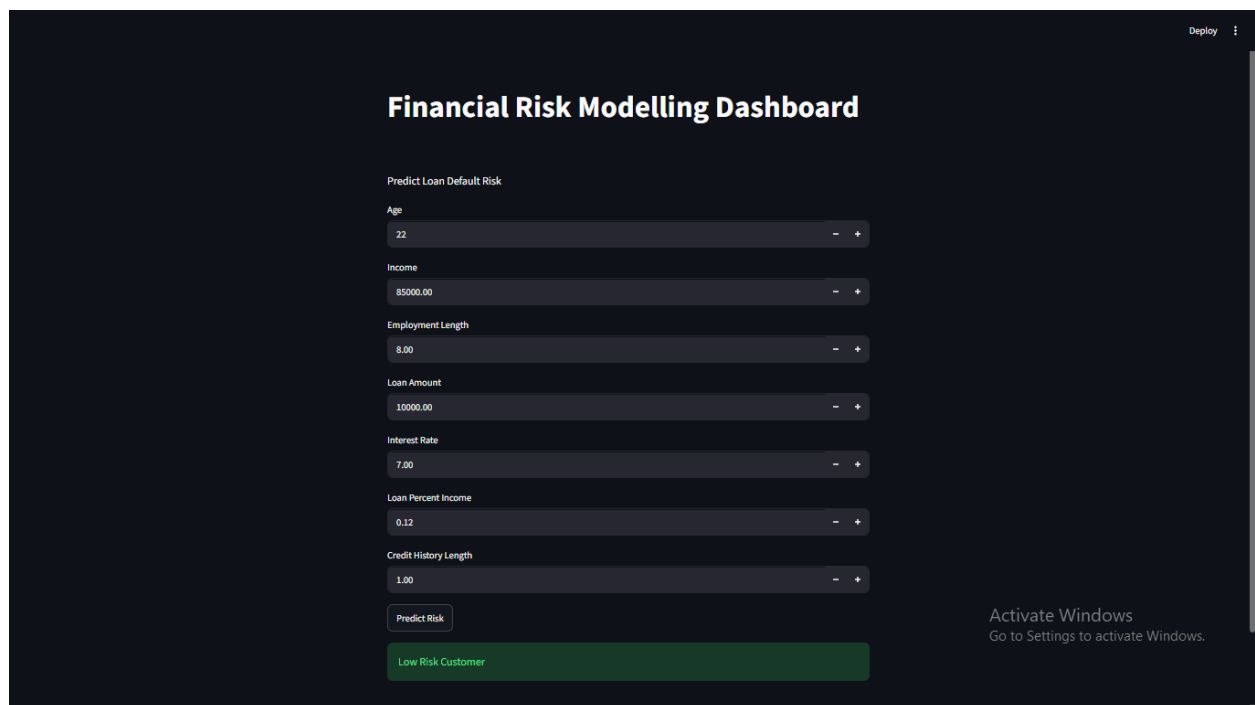
The dashboard successfully generated risk predictions and displayed the results without errors.

6.6 Dashboard Outcome

The developed dashboard successfully integrates machine learning, data preprocessing, and visualization components into a single application. It demonstrates how financial institutions can leverage artificial intelligence and data analytics to improve risk assessment processes and make informed lending decisions.

The dashboard serves as a practical implementation of Financial Risk Modelling and showcases the real-world application of machine learning in the financial sector.

Fig. 6.1 Dashboard



The screenshot displays a web-based dashboard titled "Financial Risk Modelling Dashboard". It features a dark theme with a central form for predicting loan default risk. The form includes input fields for Age (22), Income (85000.00), Employment Length (8.00), Loan Amount (10000.00), Interest Rate (7.00), Loan Percent Income (0.12), and Credit History Length (1.00). A "Predict Risk" button is located below the inputs. The result, "Low Risk Customer", is displayed in a green box at the bottom. A "Deploy" button is visible in the top right corner. An "Activate Windows" watermark is present in the bottom right corner of the dashboard area.

Parameter	Value
Age	22
Income	85000.00
Employment Length	8.00
Loan Amount	10000.00
Interest Rate	7.00
Loan Percent Income	0.12
Credit History Length	1.00

Predict Risk

Low Risk Customer

7. Conclusion

Financial Risk Modelling is an important application of data analytics and machine learning in the financial sector. The primary objective of this project was to develop a system capable of predicting the financial risk associated with loan applicants based on their personal, financial, and credit-related information.

In this project, a comprehensive workflow was followed, including data collection, data preprocessing, exploratory data analysis, visualization, machine learning model development, model evaluation, and dashboard creation. Various financial and demographic factors such as age, income, employment length, loan amount, interest rate, loan-to-income ratio, and credit history were analyzed to understand their impact on loan default risk.

Exploratory Data Analysis helped identify patterns, relationships, and trends within the dataset. Several visualizations were created to gain meaningful insights into customer financial behavior and risk factors. These insights played an important role in feature selection and model development.

To improve usability, an interactive Streamlit dashboard was developed. The dashboard allows users to enter applicant details and obtain real-time risk predictions through a simple graphical interface. The integration of machine learning models with a dashboard transformed the project into a practical application suitable for real-world use.

Overall, the project successfully achieved its objectives by developing an accurate and efficient financial risk prediction system. The findings highlight the importance of data-driven decision-making in the financial industry and demonstrate how machine learning can assist financial institutions in reducing risk, minimizing losses, and improving operational efficiency.

8. References

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4. <https://www.xerago.ai/insights>